Auditing a Released Back-Translation Evaluator for Sign Language Production: Replay Sensitivity and Public-Recipe (Non-)Reproducibility

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Abstract

Back-translation (BT) evaluators mediate sign language production rankings, but a released checkpoint may yield orderings that do not survive retraining or reference replacement. On the 641-sequence PHX-public subset of SLRTP2025, a constructed whole-donor replay probe (TN-PURE-v1) scores 23.02 sacreBLEU under the released BT evaluator versus 12.78 for the recorded test poses, a gap of **+10.24** BLEU points (95% CI **[+8.88, +11.62]**). This constructed stress test is not reproduced by fourteen independently trained evaluators reconstructed from the public recipe (all negative gaps; 95% prediction interval **[−1.94, −0.46]**), nor by 38 non-degenerate checkpoints from further training families (dev BLEU-4 4.5–10.9, all below the released evaluator's 13.38; every gap strictly negative). On the matched 461-item confidence=1 subset of Czehmann et al.'s human back-translations, reference replacement attenuates the gap from **+10.03** to **+0.95** (90.5%), a reference-frame sensitivity. The released evaluator also decodes its 7,060-item training pool at 78.8 BLEU (70.7% exact match) while retaining dev/test performance (13.38/12.78); permutation, input-shuffle with output-equivariance verification, stratified exact-match, and same-signer held-out controls exclude a pose-ignorant ID/cache lookup. Because competence-matched replication was not possible, these results evidence public-pipeline non-replication for a constructed stress test — not a causal checkpoint effect, an identified memorization mechanism, or general leaderboard-rank instability.

Keywords: sign language production, back-translation evaluation, retrieval-augmented generation, reproducibility audit, sacreBLEU, reference-frame sensitivity, PHOENIX-2014T, donor-origin contrast, evaluator non-replication, sign language recognition

1 1. Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. The present audit is framed as a *public-pipeline reproducibility audit*: whether the released SLRTP2025 evaluator and its replay-probe reversal can be reproduced from the publicly documented training recipe. We do *not* frame it as a test of checkpoint identity, nor as a direct test of the Saunders et al. Saunders et al. (2021) rank-stability proposition, which is conditioned on similar model configurations: every checkpoint we can construct from the public recipe is less competent than the released evaluator (dev BLEU-4 4.5–10.9 vs. 13.38), so competence and checkpoint identity are confounded and the non-reproduction result is scoped to the public recipe.

Prior work shows BT scores can be modest for recorded signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2025); sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai (2025); Jiang et al. (2025), yet BT remains central to SLRTP2025 Walsh et al. (2025). An unresolved question is whether a retrieval–reference stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by a retrieval–reference reversal on PHX-public: under the original evaluator, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU (0–100 scale) versus 12.78 for the recorded test poses — which neither establishes better signing nor invalidates BT evaluation in general, but asks whether the reversal persists across evaluator checkpoints or depends on properties of the original evaluator.

We address this question and report three findings. *First, the reversal does not replicate across independently trained evaluators reconstructed from the public recipe*: fourteen same-architecture reconstructions yield a seed-level PURE–REC gap prediction interval of $[-1.94, -0.46]$ BLEU points, and 38 non-degenerate checkpoints from further training families (distillation, config-faithful, step-faithful, rescue) extend the negative range to $[-2.01, -0.21]$. *Second, the reversal is partly a property of the reference frame*: on the matched 461-item confidence=1 subset of public human sign-to-text back-translations Czehmann et al. (2025), the gap attenuates from +10.03 to +0.95 BLEU points (90.5%), an asymmetry that is inconsistent with simple floor-effect compression. *Third, the released evaluator exhibits a readout-with-generalization signature that no constructible checkpoint reproduces*: it decodes its 7,060-item training pool at 78.8 BLEU (70.7% exact-match) while retaining dev/test performance of 13.38/12.78, and a weight-perturbation and fine-tuning sweep preserves the reversal near the released weights before it collapses in lockstep with competence. The three findings are developed in Section 4 through four lines of evidence (conditioning controls, multi-checkpoint replications, reference replacement, and five diagnostic views); evaluator-aware oracle analyses are relegated to SI Sup. H because their unmasked advantage does not survive reference-overlap exclusion (audit design in SI Fig. S4).

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator's training pool, so TN-PURE-v1 isolates a risk that production systems face in diluted form; methodologically, the audit contributes a template — provenance table, checkpoint registry, evaluator-replication controls — that other benchmarks can adopt for public-recipe reproducibility testing.

2 2. Related Work

Learned evaluation in sign language production.

Neural SLP often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020); BT was proposed as an SLP evaluation protocol by Saunders et al. Saunders et al. (2021), who argued it shifts absolute scores but not relative rankings between similar configurations — a proposition our audit does not directly test (its similarity condition is unmet in our family) but re-examines for a released evaluator on which a retrieval-probe reversal appears. SignBLEU Kim et al. (2024), SiLVERScore Imai (2025), and pose-native measures offer complementary views, but BT BLEU remains central to SLRTP2025 Walsh et al. (2025).

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025), and SLRTP2025's leading entry is retrieval-based, making donor copying pertinent Walsh et al. (2025). Memorization results in retrieval-augmented language modeling show strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020); a shortcut under one learned evaluator does not show the ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2025) show BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low ground-truth BT baseline; Jiang et al. Jiang et al. (2025) recommend BT likelihood over decoded BLEU; and model-based generation metrics are further criticized by Yazdani et al. (2025); Cory et al. (2025); He et al. (2024). Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2025) released human back-translations and documented reference information loss, Alkain et al. Alkain et al. (2026) showed the highest train–test text overlap among tested SLT corpora, and Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

Table 1: Three-way data provenance for PHX-public. The released evaluator's training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs share the reason `absent_from_slrtp2025_released_pose_records` (SI Sup. A). SHA-256 hashes verify file integrity.

Split source	Train	Dev	Test
Official PHOENIX-2014T	7,096	519	642
Released SLRTP2025 pose (`.pt`)	7,060	515	641
Released evaluator `config.yaml` manifest	7,060	515	641
Missing from official	36	4	1

3 3. Methods

3.1 3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint and $o_{i,n}$ the pose sequence for output condition i and item n; E_c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^{N})$, where y_n is the reference text and M is the corpus-level sacreBLEU functional unless stated otherwise. We call a result *evaluator-pipeline-sensitive* when a substantive pairwise ordering changes across independently trained checkpoints under a fixed output set. The fixed primary set comprises recorded test poses (REC), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval–generator compositions; oracle-gloss variants are diagnostic controls.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark whose released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). Table 1 gives the three-way provenance: the official PHOENIX-2014T split, the released pose materialization, and the released evaluator's actual training manifest (confirmed via `config.yaml`, whose manifest matches the shipped `.pt` tensors). Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. L); the protocol summary is in SI Table S28.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution; a three-path regression confirms no double subsampling and that the official starter-bundle Progressive Transformer reproduces the official 1.5942 BLEU under our pipeline (details in SI Sup. F). PHX-public uses α=0.88 (selected on dev); the supplementary splits use α=0.5.

3.2 3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, selecting the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic;

random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ; gloss-based exclusion is therefore an offline diagnostic, not a deployable inference procedure.

Single canonical materialization. Every number in this paper is generated from one canonical donor registry built by the deterministic §3.2 algorithm (exact-normalized-text exclusion applied), whose SHA-256 is `9170a53026ab3263b451a3632a9e02-318745acabf12f0b679921477afbefa301`; the artifact regenerates the registry and all headline numbers with a single command. An earlier pre-exclusion materialization (PURE 23.79, gap +11.01) survives only as a sensitivity analysis in SI Sup. N: 610/641 items (95.2%) are identical across materializations, and a 14-reconstruction comparison gives mean $|\Delta\text{gap}|$ = 0.65 BLEU (max 1.45), so qualitative conclusions are unchanged.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with [::2] subsampling to 12.5 fps, preserving the donor's original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α-blending of upper-body joints (α=0.88; full joint-set definitions in the artifact). The matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the retrieval advantage co-occurs with donor origin, we construct frozen probe systems from the same 641 queries: **UNSEEN-PURE-v1** retrieves from the 640 other test poses; **CROSSFIT-PURE-A/B-v1** split queries by hash into two folds, each retrieving from the other fold; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ±0.1 Jaccard of each UNSEEN donor (622/641 admit one); **SEEN-RAND640-MATCHED-v1** applies matched selection inside frozen 640-donor sub-pools. SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry; these are constructed probes, not deployable generators (construction rules in the artifact).

3.3 3.3 Evaluator checkpoint family

The primary inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and six *recipe-conditioned reconstruction* runs trained from scratch with the documented architecture on a common frozen partition (primary seeds 101–606; eight extension seeds 707–1405 use the identical protocol). The original bundle's `validations.txt` logs 202 validation points over 2,828 optimizer steps (≈102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; the trajectory is analyzed in §4.2.1, and intermediate checkpoints are not published. We use two reconstruction protocols — *paper-derived* (14 runs, legacy script, 300 epochs, NLL selection) and *released-config* (4 runs, up to 3,000 epochs, decoded-BLEU early stopping) — sharing the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), [:: 2] subsampling, Adam (1e-3, wd .001), batch size 256, and plateau ×0.8 LR schedule (full provenance in SI). In total 70 runs across 13 families (registry in SI Sup. D); 43 were decoded on PHX-public under the canonical §3.2 donor registry, of which five are degenerate (three α=1.0 distillation students with empty hypotheses, two smallest ladder fractions with BLEU ≈ 0) and are excluded, so the non-replication count is 38 non-degenerate checkpoints.

Readout-with-generalization signature (post-hoc operational definition). A checkpoint exhibits the training-pool readout signature if it has both (i) training-pool free-decode BLEU ≥ 50 *or* exact-match $\geq 50\%$, and (ii) dev BLEU-4 ≥ 8.0; the thresholds were defined after observing the released evaluator (78.8 BLEU / 70.7% EM, dev 13.38) vs. the reconstruction family (max 11.5 train / 9.9 dev, no simultaneous satisfaction), and are operational heuristics for this case study, not validated criteria.

3.4 3.4 Metrics and uncertainty

The recorded test poses (REC) are decoded by each BT recognizer; we avoid the abbreviation "ground truth" because the recorded poses are inputs to a learned measurement model, not a semantic gold standard. We report corpus BLEU-4 on the 0–100 sacreBLEU scale as the primary metric Post (2018); sentence-level statistics (pass-through, slot-F1) are reported separately on the 0–1 scale. BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020) (`bert-base-multilingual-cased`, no baseline rescaling, F1), and WER (jiwer 3.1.0 normalized) provide complementary decoded-text controls; BT likelihood and pose-DTWp provide non-decoded controls; a non-standard BLEURT substitute used only for a secondary reference-sensitivity check is documented, with its validation against canonical BLEURT-20 and its known bias, in SI Sup. O.

Uncertainty is quantified with paired sequence-level bootstrap (10,000 resamples, seed 42, percentile 95% CIs) over per-item n-gram sufficient statistics, re-aggregated into a corpus BLEU per resample. Because queries reuse pose donors (565 unique donors for 641 queries), donor-cluster and query$\times$donor two-way cluster bootstraps complement the item bootstrap (SI Sup. G). A pairwise difference is called detectable when its interval excludes zero. A descriptive seed-level 95% Student-t prediction interval is reported for the recipe-conditioned reconstruction population (the eight extension seeds enter only this interval, never the six primary comparisons). The single primary estimand is the PURE–REC decoded-sacreBLEU gap under the released evaluator; all other contrasts are secondary and are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

4 4. Results

4.1 4.1 Retrieval reverses the reference ranking

Under the original SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses — a retrieval–recorded-poses gap of +10.24 BLEU points (95% CI $[+8.88, +11.62]$; 10,000 sequence-level resamples). This fixed-checkpoint reversal motivates the analysis but does not indicate superior sign production.

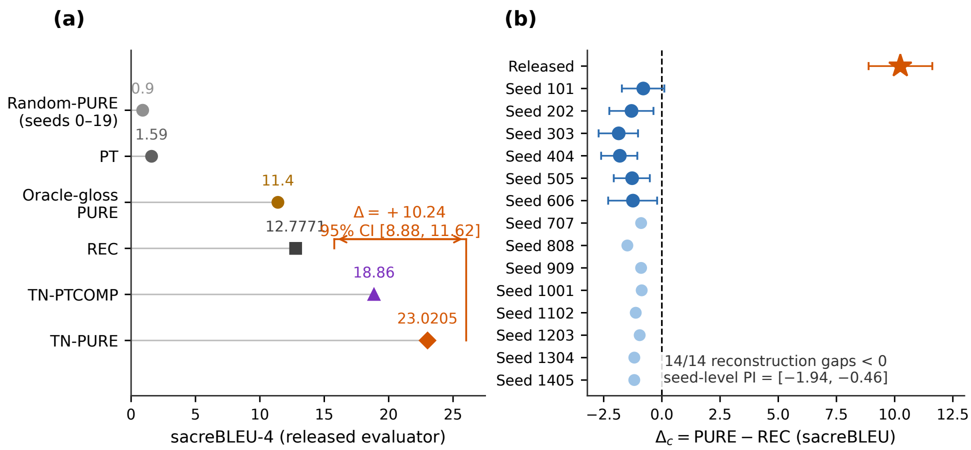

Fig. 1: Headline reversal under the released evaluator and its absence across recipe reconstructions. (a) System scores under the released evaluator (sacreBLEU-4, canonical 641-query protocol): TN-PURE 23.02 vs. REC 12.78 ($\Delta = +10.24$, 95% CI $[8.88, 11.62]$), with TN-PTCOMP 18.86, PT 1.59, oracle-gloss PURE 11.4, and random-donor PURE 0.90 (seeds 0–19, ± 1 SD). (b) Per-checkpoint PURE−REC gaps: released evaluator (+10.24) vs. fourteen recipe reconstructions, all negative (paired-item bootstrap 95% CIs; seed-level 95% PI $[-1.94, -0.46]$). All reconstructions are less competent than the released evaluator; the figure demonstrates public-pipeline non-replication, not an isolated causal checkpoint effect.

Controls localized the observation to source-conditioned donor retrieval (Table 2): pure donor copy exceeded the oracle-gloss composed variant (11.8), canonical PT-composed (18.86), PT baseline (1.59), and 20-seed random-donor baselines (0.90 PURE, 0.77 COMP), while oracle-gloss retrieval (leakage) did not exceed REC (11.4–11.8 vs 12.78). Non-decoded controls do not reproduce the reversal: teacher-forced NLL gives REC 3.060 vs PURE 3.479, COMP 3.411, PT 4.074, and normalized pose-DTW shows REC 0 by construction vs PT/PURE/random 0.00678–0.00830. Complementary text metrics agree on the same hypotheses (SI Table S22): sacreBLEU-4 +10.24, BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER −13.90, BERTScore-F1 +0.066; we therefore describe the observation as a *decoded-text-metric reversal* with sacreBLEU-4 as the primary decoded metric.

Decoding-parameter sensitivity.

Re-decoding across beam size $\{1, 3, 5\}$, length penalty $\{-1, 0\}$, and max length $\{30, 50\}$ leaves the PURE–REC gap at +9.9/+10.2/+10.8, so the reversal is not a decoding artifact; length penalty and max length are negligible.

Pure donor copy outperformed composition for every source-conditioned retriever, and source-text Jaccard was highest (23.02, vs LaBSE 19.2, multi-SBERT 16.5): the reversal needs no oracle-gloss leakage, and pure copy — not composition — is its strongest form.

Table 2: PHX-public conditioning and construction controls (641 sequences). Columns distinguish whole-donor copy from PT-scaffold composition; all rows share the query set, donor pool, evaluator, and sacreBLEU protocol. Semantic retrievers (LaBSE, multi-SBERT) lie between the canonical retriever and oracle gloss (artifact).

Conditioning	Pure donor copy	PT+retrieval composed
Source text (German Jaccard)	**23.02**	18.86
Oracle gloss (input leakage)	11.4	11.8
Random donor (seeds 0–19)	0.90 (SD 0.24)	0.77 (SD 0.19)

4.2 4.2 Independent reconstructions do not reproduce the reversal

TN-PURE-v1 yields an original-evaluator gap of +10.24 BLEU points (95% CI $[+8.88, +11.62]$) and TN-PTCOMP-v1 +6.09 $[+4.76, +7.41]$; across all fourteen reconstructions both gaps are negative (PURE range $[-1.85, -0.80]$; seed-level 95% prediction interval $[-1.94, -0.46]$, mean -1.20, SD 0.33, 0/14 positive), and for the six primary runs original-minus-run difference-in-differences estimates range from +10.89 to +11.99, every interval excluding zero.

4.2.1 4.2.1 Recipe-conditioned evaluator reconstructions

All fourteen reconstructions share frozen manifests, architecture, hyperparameters, and selection rule; seeds 101–1405 vary initialization and data order (Table 3), and all evaluators are decoded with the canonical §3.2 donor registry (exact-normalized-text exclusion applied).

Table 3: Canonical fixed-output evaluation on the released 641-pose PHX-public subset, decoded with the §3.2 donor registry. PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1. Gaps are relative to local REC (BLEU points). Per-cell values in the artifact.

Evaluator	REC	PURE	Gap	COMP	Gap
Original	12.78	23.02	**+10.24**	18.86	+6.09
Seed 101	9.24	8.45	−0.80	7.92	−0.65
Seed 202	8.22	6.92	−1.30	7.54	−1.28
Seed 303	9.10	7.25	−1.85	8.71	−0.96
Seed 404	9.52	7.71	−1.81	6.38	−0.81
Seed 505	9.71	8.44	−1.28	7.97	−1.08
Seed 606	7.55	6.31	−1.24	7.93	−0.16
Extension 707–1405 (range)	8.05–9.87	6.86–8.97	−1.48 to −0.87	6.90–8.83	−1.03 to −0.44

Competence is assessed on the released 515-sequence development subset, never on PHX-public. Original obtains dev BLEU-4 13.38; under the uniform full-pool beam-3 free-decode protocol the 14 reconstructions obtain dev BLEU-4 6.5–9.4, and their selected validation NLL (2.641–2.712) is lower than Original's teacher-forced dev NLL (3.235), so likelihood and decoded recognition competence are not interchangeable. Consequently, competence and unavailable original-training details remain confounded with evaluator identity; the experiment identifies evaluator-pipeline sensitivity, not a causal checkpoint-identity effect.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both $|\Delta\text{dev BLEU-4}| \leq 1.0$ and $|\Delta\text{dev WER}| \leq 3.0$ (released dev values 13.38 / 83.37). None of the 52 non-distillation trained checkpoints passes either half (dev BLEU-4 short by 3.4–8.9; dev WER exceeds the 86.37 bound by 0.08–13.3), and the rescue search (batch size, dropout, weight decay, label smoothing, longer budget; SI Table S29) also fails: its best variant (weight-decay 0, seed 202) yields REC 9.88 / PURE 8.29 (gap −1.59). The gate conclusion is threshold-insensitive: across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid, 0 of 52 pass.

What the released training log establishes.

The bundle's `validations.txt` logs dev BLEU-4 reaching 13.38 at step 1,820 (Fig. 2a), while all constructible reconstructions plateau at 8–10 by epoch 50–100; the training dynamic behind this slope is not explained by public artifacts. Input degradation (temporal downsampling) compresses both scores but does not eliminate the reversal (at k=8: REC 8.75, PURE 11.69, gap +2.94; SI Sup. F).

Cross-metric consistency across checkpoints.

SI Table S24 reports the gap on the same fixed output under each of the seven primary evaluators across metrics. Under the original, the gap is positive and large for all decoded-text metrics, while teacher-forced NLL is more negative for PURE than REC (−0.419 sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2025). Under the six reconstructions the reversal disappears: all chrF and ROUGE-L gaps are negative, all WER gaps favor REC, and sacreBLEU-4 gaps are all negative (range $[-1.85, -0.80]$): the reversal is a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4.

Rank stability (see SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank REC first, with moderate cross-checkpoint agreement (mean τ_b=0.44, ρ=0.57).

Reference-frame sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2025) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical

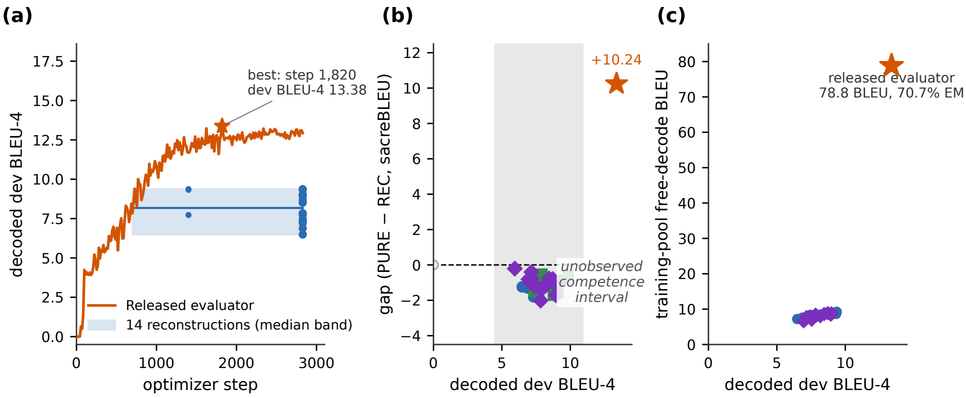

Fig. 2: Decoded competence and training-pool readout separate the released evaluator from every constructible checkpoint. (a) Dev-BLEU-4 training trajectories on a common optimizer-step axis: the released evaluator (orange) reaches 13.38 at step 1,820 of 2,828 (archived validation log, 202 logged points); fourteen reconstructions form a median band (blue), plus uniformly decoded epoch-25/50 points. (b) Canonical PURE−REC gap vs. decoded dev BLEU-4 by family: reconstructions (circles), config-faithful/step-faithful/confirmation/long-schedule (squares), rescue (triangle), distillation students (diamonds), released evaluator (star); the gray band marks the observed 4.5–10.9 range and the (10.9, 13.38) interval is unobserved; degenerate checkpoints are hollow gray points. (c) Full-training-pool free-decode readout (beam-3, full 7,060 poses) vs. decoded dev BLEU-4: every constructible checkpoint reads out 5.9–9.4 BLEU, while the released evaluator reads out 78.8 BLEU (70.7% exact match) at dev 13.38. Families are descriptive; checkpoints are not IID replicates.

errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train–test text overlap across sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score REC 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score REC 11.93 / PURE 22.20 (gap +10.3).

Czehmann et al. compared original- and human-reference conditions on the same confidence=1 subset (462 of 642 items); we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.03 BLEU points (REC 14.94 / PURE 24.98) and the human-reference gap is +0.95 (REC 7.35 / PURE 8.30): a paired attenuation of 9.08 BLEU points (90.5%, bootstrap CI $[7.44, 10.86]$; SI Table S31; SI Fig. S5; per-item paired view in SI Fig. S2), and the residual gap is not distinguishable from zero (paired item CI $[-0.13, 1.99]$). Teacher-forced BT likelihood does not reproduce the reversal under either reference (REC NLL 3.060 vs PURE 3.479 on original refs; 3.41 vs 3.71 on human refs) Jiang et al. (2025). Scoring against both references jointly (closest-length convention) leaves the gap at +11.50 — *larger* than original: TN-PURE donors are selected by lexical overlap with the source sentence, and the original reference is a speech transcription of that same sentence, so adding it injects a second lexically coupled target the replay decode can match; the gap inherits the lexical coupling rather than averaging it away. The reversal is thus specific to reference frames containing the original speech-transcribed register. The full-641 comparison (+10.24 $\rightarrow$ +0.86, 91.6%) mixes reference replacement with floor effects from low-confidence items. This is a *descriptive contrast of two corpus gaps*, not a causal attribution: the attenuation cannot distinguish reference-text defects, paraphrase/translationese shift, BLEU floor effects, disruption of the lexical coupling, or evaluator training-distribution alignment. Under all six reconstruction checkpoints the human-reference gap is negative (-0.15 to -0.65), and the asymmetry is consistent across metrics (chrF 82.2%: +12.73 $\rightarrow$ +2.27; BLEU-1 84.0%: +20.01 $\rightarrow$ +3.19; SI Table S23), including a BLEURT-substitute check agreeing with Czehmann et al. (SI Sup. O). It quantifies how decoded-text scores shift when the reference frame changes, not whether the replay donor motion conveys target-equivalent meaning: the human back-translations are sign-conditioned paraphrases of the recorded test poses, not judgments of the replay donors, so the semantic adequacy of TN-PURE-v1 is left unassessed.

A natural concern is symmetric floor compression. Switching from original to human references, REC drops 7.59 BLEU-4 points (50.8%) but PURE drops 16.68 (66.8%) — a larger fraction (ratio 1.31$\times$), and the asymmetry is even larger under BLEU-1 and chrF (SI Table S23); simple symmetric compression is insufficient, though the asymmetry does not discriminate between reference-text defects and training-distribution alignment.

4.3 4.3 Donor-origin contrast

If the released-evaluator reversal requires exposure to training-pool donors, retrieval restricted to unseen poses should remove or invert it: under the original evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.51 sacreBLEU vs. REC 12.78 (gap -4.27), a donor-pool substitution estimand of +14.51 BLEU points (SI Table S30). This is a *donor-pool substitution effect*, not a pure exposure effect: SEEN and UNSEEN differ simultaneously in pool size (7,060 vs 640), median donor–reference Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure. Cross-fit controls reproduce the direction (CROSSFIT pooled -3.8), and all reconstruction pool-origin contrasts lie in $[-0.15, +1.22]$.

Path-dependent decomposition and balanced factorial.

Telescoping-path decomposition (SI Sup. E) gives origin estimates of +5.8 to +8.4 BLEU across admissible orderings, because size and selection interact. The better-balanced 605-query factorial (matching seen/unseen donors at high/low LaBSE similarity) estimates the pool-origin main effect at +2.08 $[+1.56, +2.57]$, similarity at +6.96 $[+5.93, +8.08]$, and their interaction at +3.70 $[+2.63, +4.79]$ (all Holm p=0.0003); across reconstructions the seen main effect shrinks to +0.51 and the interaction is null (-0.39). Because seen donors remain more similar than unseen within strata (SMD +0.4), this is a train-pool vs test-pool provenance contrast, not an identified training-exposure effect.

4.4 4.4 Competitive-system context

Because the audit's stress-test set is constructed rather than competitive, the public SLRTP2025 leaderboard offers external context: computed under the same released evaluator, its "Ground Truth" row reproduces our canonical REC numbers exactly (BLEU-4 12.78, BLEU-1 34.40, WER 85.77 Walsh et al. (2025); SI Table S25). The retrieval-based winner USTC-MoE sits roughly even with REC on the lexical BT metrics (BLEU-4 −0.72, BLEU-1 +0.45, chrF +2.24; WER 93.49 vs 85.77) while its DTW-MJE (0.0448) and duration ratio (1.438) indicate considerable pose-level deviation from recorded signing — a single-evaluator co-occurrence that does not establish that the released evaluator favors retrieval entries; resolving that question would require re-decoding fixed competition outputs under multiple evaluators.

Scope limit and cross-evaluator re-decoding.

The public artifacts include only the Progressive Transformer baseline output (`PT_baseline_test.pt`, evaluated as PT-v1); per-team pose outputs are not archived, so competition systems cannot be re-decoded under our reconstruction family, and whether USTC-MoE's roughly even lexical-BT scores survive across evaluators remains an open empirical question. As a benchmark-organizer responsibility, archiving fixed per-team outputs alongside public scores would make rank-stability audits feasible for third parties.

Released-checkpoint local-robustness checks (experiments A1–A2, SI Sup. F).

Two perturbation designs test whether the reversal is a fragile singular point versus a locally robust property near the released checkpoint. Experiment A1 adds i.i.d. Gaussian weight noise at increasing scale σ: the reversal is preserved within ± 0.7 of $+10.24$ for $\sigma \leq 3 \times 10^{-3}$ (where dev BLEU-4 stays above 12.7) and collapses in lockstep with competence at $\sigma \geq 10^{-2}$. Experiment A2 fine-tunes the released checkpoint with three seeds at small learning rates (5e-6–3e-5) for 60 epochs, moving weights along real gradient directions rather than isotropic noise; all fine-tuned variants preserve the reversal. Both designs confirm that the reversal is not a fragile identity-level artifact in the released checkpoint's parameter neighborhood, but they probe only the vicinity of the released weights — they do not demonstrate that an independently trained, competence-matched model would exhibit the reversal, nor do they narrow the unobserved competence interval above the family ceiling (10.9). Full sweep, epoch-decoupling search, and figures in SI Sup. F.

Per-item gap decomposition (experiment C, SI Sup. K).

An OLS regression ($R^2 = 0.33$, $n = 641$) attributes the largest share of per-item gap variance to high-overlap-train count, template density, and donor–reference Levenshtein distance; signer identity contributes $\leq 1\%$. Full model, coefficient table, and figure in SI Sup. K.

4.5 4.5 What distinguishes the released evaluator

The first finding (negative gaps across reconstructions) raises a positive question: what property of the released evaluator do the reconstructions lack? We approach it through five diagnostic views that triangulate on a single signature rather than independently confirming it (Fig. 2b,c; Table 4; full version in SI Sup. D), reporting them as descriptive triangulation, not five independent tests.

Table 4: Summary of the five diagnostic views. The released evaluator sits far outside the family range on all five views. Per-view family composition in SI Sup. D.

Diagnostic view	Family n	Released evaluator	Family range	Gap to family max
(i) Gap–competence	43	+10.24 at dev 13.38	$[-2.01, 0.00]$ at dev 4.5–10.9	+10.24
(ii) Pass-through ratio	24	3.09 (sentence)	1.05–1.31	+1.78
(iii) Train-pool readout BLEU	27	78.8 (70.7% EM)	6.6–9.3 (1–2% EM)	+69.5
(iv) Frame-reversal loss	1	−16.1 (PURE)	−4.3 (wd0 rescue)	|11.8|
(v) Competence gate	52	13.38	4.5–10.02 (max)	+3.36

Readout* with *generalization: the distinctive combination.

The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match — strong readout — while retaining non-trivial dev/test performance (13.38/12.78). Under the identical full-pool, beam-3 free-decode protocol, the constructible family stays far below: 14 reconstruction seeds span train BLEU 7.2–9.3 (EM 1.5–2.3%) at dev BLEU 6.5–9.4, and 12 non-degenerate distillation students span train BLEU 6.6–8.9 (EM 1.3–1.9%) at dev BLEU 7.0–9.0 (training-time dev BLEU-4 under the original selection protocol reaches 10.9; SI Table S26, SI Sup. E). None of the 27 non-degenerate constructible checkpoints approaches 78.8/70.7%, and the released evaluator's donor-content pass-through is a 2.4× outlier (ratio 3.09 vs. family 1.05–1.31).

Leakage sanity checks for the train-pool readout.

The 78.8/70.7% readout is not a trivial implementation artifact: five controls (SI Sup. M) rule out four shortcut classes — teacher forcing (code-path audit confirms autoregressive beam search), output–reference misalignment (label permutation collapses BLEU from 78.6 to ∼1.0), pose-ignorant ID/cache lookup (input-side pose permutation with verified output equivariance collapses BLEU to <1; a same-signer held-out control reaches only 12.19 BLEU / 3.2% EM on unseen poses), and narrow-text memorization (high exact-match across all 9 signers and unique texts). The checks do not identify the training mechanism but are consistent with pose-conditioned training-pool exposure.

Knowledge-distillation competence ladder.

To probe the unobserved competence interval above the reconstruction ceiling (dev BLEU-4 10.0), we trained 15 distillation students from the released evaluator as teacher (five strengths $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$, three seeds, T=2). All twelve α<1.0 students have negative PURE–REC gaps (−2.01 to −0.77) with dev BLEU-4 up to 10.9 and readout 6.6–8.9 BLEU / 1.3–1.9% EM; the three α=1.0 students degenerate into empty or near-empty outputs and do not constitute a valid transfer test (SI Table S26). Under the tested procedure, the readout signature therefore does not transfer.

Competence extrapolation (exploratory).

Three fits (OLS linear, OLS quadratic, GP with RBF + white kernel) over the 43 checkpoints spanning dev BLEU-4 $[4.5, 10.9]$ all predict negative gaps at 13.38, with substantial model sensitivity (OLS quadratic flips from −13.5 to +1.3 when distillation is removed; GP upper bound stays below +2.0 across leave-one-family-out folds). We treat this as an exploratory visual guide (SI Sup. K), not evidence that competence alone cannot explain the reversal, since non-linear behavior inside the unobserved $[10.9, 12.7]$ interval cannot be excluded.

Synthesis.

Across the five diagnostic views, the released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain performance (13.38/12.78). No reconstruction or distilled student reproduces either the +10.24 gap or this signature.

4.6 4.6 Checkpoint-targeted oracle stress tests (SI)

We treat evaluator-aware donor selection as a checkpoint-targeted oracle stress test, not a deployable SLP method: the unmasked diagonal advantage does not survive reference-overlap exclusion, and the white-box search budget (candidate-pool size) controls whether any decoded-BLEU margin is detectable. The full cross-checkpoint transfer matrix, candidate-pool sweep, and selector/evaluator-label permutation test are in SI Sup. H, which they share with the main estimands by design.

5 5. Discussion

The audit yields a broad negative result and a descriptive positive characterization. The negative result is carried by the fourteen same-recipe reconstruction seeds (shared architecture, tokenizer, manifest, selection rule; seeds 101–1405), whose Student-t prediction interval for the PURE–REC gap is $[-1.94, -0.46]$ BLEU points; beyond them, 38 non-degenerate checkpoints from heterogeneous families (distillation, config-faithful, step-faithful, rescue, confirmation; all decoded with the canonical donor registry) have strictly negative gaps (range $[-2.01, -0.21]$), reported as exploratory dose–response data, not IID replicates (SI Sup. D). The positive characterization is that the released evaluator carries a high training-pool readout combined with retained in-domain performance that no public artifact in our family exhibits (§4.5). Because all constructible checkpoints are less competent than the released evaluator (family max dev BLEU-4 10.9 vs. 13.38), this non-reproduction is observed under a competence confound. Two local-robustness checks on the released checkpoint (Gaussian weight noise and fine-tuning from released weights, SI Sup. F) confirm the reversal is not a fragile identity-level artifact nearby, but they probe only the vicinity of the released weights and do not narrow the unobserved competence interval above the family ceiling (exploratory OLS/GP extrapolation in SI Sup. K, visualization only). Because Saunders et al. Saunders et al. (2021) conditioned rank stability on similar model configurations, and our family is additionally competence-unmatched, our non-reproduction result is scoped to the public training recipe and does not constitute a strong test of their proposition.

On the matched 461-item confidence=1 subset, human-reference substitution attenuates the gap by 90.5% (+10.03 $\rightarrow$ +0.95, paired bootstrap CI $[7.44, 10.86]$); the asymmetric PURE-vs-REC drop ratio (1.31$\times$ BLEU-4; 1.29$\times$ full-641) is inconsistent with symmetric compression but does not discriminate between reference-text defects and training-distribution alignment. Czehmann et al. Czehmann et al. (2025) already showed that reference replacement changes BLEU scores and rankings on PHOENIX-2014T; our contribution is applying the matched-subset protocol to the released evaluator and quantifying paired attenuation. Because TN-PURE-v1 reuses pose donors across queries (641 queries, 565 unique donors, Gini 0.11), item-level resampling overstates the effective sample size, so we also resample at the donor level; the headline gap is stable under all cluster-aware resamplings (SI Sup. G) — donor-cluster, query$\times$donor two-way, signer, broadcast-date, signer$\times$show — all consistent with the query-level CI $[8.88, 11.62]$.

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to public-recipe reproducibility: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2025), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2025). Our reversal is a *decoded-sacreBLEU* reversal — teacher-forced BT likelihood does not reproduce it (REC NLL 3.060 vs PURE 3.479) — and whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025).

Actionable recommendations for benchmark organizers.

The audit yields three transferable lessons, scoped to the constructed replay-probe stress case on PHX-public rather than to general SLP leaderboard stability.

Lesson 1: a released evaluator is a research artifact, not an infrastructure component. The released `config.yaml` alone was insufficient for 70 same-recipe runs to reach competence parity, and the readout-with-generalization signature could not be localized to a training stage because intermediate checkpoints are unpublished. Benchmarks should release (i) the exact training command and seeds, (ii) data pipeline scripts beyond the manifest, (iii) a SHA-256 manifest of training tensors, (iv) intermediate checkpoints at least every 100 epochs, and (v) the fixed per-team pose outputs alongside the public scores; the current release provides only the Progressive Transformer baseline output, so rank-stability audits across independently trained evaluators cannot be reproduced by third parties on real competition systems.

Lesson 2: retrieval-based entries require shortcut-robustness certification. A donor-exclusion sweep is cheap, deterministic, and exposes the retrieval advantage; leaderboard entries that retrieve from the training pool should attach $\tau \in \{0.3, 0.5\}$ donor-exclusion results as a shortcut-robustness certificate. Two descriptive probes further separate genuine signing from pipeline artifacts: the training-pool free-decode BLEU $\div$ dev free-decode BLEU ratio (released: 6.1; reconstructions: 1.1–1.4) flags anomalous training-pool alignment, and the input-side pose permutation with output-equivariance verification (SI Sup. M) excludes a pose-ignorant ID/cache lookup; the released evaluator passes the latter but fails the former by a wide margin.

Lesson 3: reference validity is part of evaluation validity. The 90.5% attenuation under Czehmann et al. Czehmann et al. (2025) back-translations shows that rankings against a single speech-transcribed reference can misrepresent sign-language quality. Benchmarks should adopt human back-translations as a secondary reference set, follow the matched-subset protocol (confidence=1 items), and flag systems whose gap diverges sharply between the two; the cost is one reference file and a paired bootstrap, the benefit is detection of reference-coupling artifacts that single-reference BLEU cannot surface.

Limitations.

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2025); Alkain et al. (2026). The fixed nine-system stress-test set comprises replay, composition, weak generation, and random controls rather than competitive SLP systems; the public leaderboard (§4.4) offers only single-evaluator context, and because per-team pose outputs are not archived, those systems cannot be re-decoded under the reconstruction family — no causal claim about evaluator retrieval-favoring follows. All 67 non-holdout checkpoints are less competent than the released evaluator (family max dev BLEU-4 10.9 vs. 13.38); the original training history is unpublished, so competence-matched replication from the public recipe was not possible, and local-robustness checks (Gaussian weight noise and fine-tuning from the released checkpoint, SI Sup. F) confirm the reversal is not a fragile singular point but do not narrow the unobserved competence interval. Saunders et al. Saunders et al. (2021) limited their rank-stability claim to similar configurations, which our competence-unmatched family does not directly test. The mechanism is not identifiable from public artifacts: neither recipe-conditioned reconstruction, rescue sweeps, config-faithful retraining, nor any tested distillation procedure transfers the property; the donor-pool substitution decomposition is path-dependent, and pool-origin contrasts are descriptive associations, not identified training-exposure effects. The human back-translations are a single-annotator derivative resource without second-annotator adjudication. Finally, the reversal is a decoded-BT-text-metric reversal, not a sign-language quality reversal:

without target-conditioned DGS signer judgment or a validated sign-native pose metric, we cannot determine whether the retrieved donor motion conveys target-equivalent meaning, and we explicitly do not claim that donor motion fails to express the target, nor that the reversal indicates dataset contamination or pose memorization.

6 6. Conclusion

This audit reports three findings on a +10.24 BLEU-point retrieval–recorded-poses reversal under a released BT evaluator on PHX-public. The reversal does not replicate across 38 non-degenerate checkpoints from independent training families (14 same-recipe reconstructions plus 24 heterogeneous; range $[-2.01, -0.21]$, single canonical donor registry), attenuates by 90.5% on a matched confidence=1 subset under independent human back-translations, and co-occurs with a training-pool readout signature (78.8 BLEU / 70.7% EM, retained dev/test 13.38/12.78) that no constructible checkpoint exhibits, with permutation controls ruling out a pose-ignorant ID/cache lookup.

The scope is correspondingly bounded. Local-robustness checks (Gaussian weight noise and fine-tuning from the released checkpoint) confirm the reversal is not a fragile singular point, but they probe only the vicinity of the released weights and do not narrow the unobserved interval between the family ceiling (10.9) and the released checkpoint (13.38); the training log shows validation PPL rising from 14.84 at step 546 to 25.42 at the best-BLEU step 1,820, so decoded-BLEU selection chose a checkpoint above the validation-NLL minimum. Because competence-matched replication was not possible from public artifacts, the findings evidence public-pipeline non-replication for a constructed stress test, not a causal checkpoint effect, an identified memorization mechanism, or general leaderboard-rank instability.

Compliance with Ethical Standards

Conflict of interest.

The authors declare no conflict of interest.

Ethical approval.

All experiments use publicly released datasets (PHOENIX-2014T, CSL-Daily, SLRTP2025) and Czehmann et al.'s human back-translations. No animal research was conducted.

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Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the anonymous artifact at https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md; a persistent DOI-bearing release will be deposited upon acceptance. Upstream datasets are used under their published licences (PHOENIX-2014T and the human back-translations under CC BY-NC-SA 4.0; CSL-Daily and SLRTP2025 under their academic licences); pose tensors may retain signer identity, so we release no per-signer raw poses.

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (`supplementary.pdf`) and in the anonymous artifact. Each appendix is annotated with the question it answers.

- Sup. A: Missing Test ID Sensitivity — does the absent test item change the headline gap? (`sec:appendix_missing_id`)
- Sup. B: Per-seed Extension Cells 707–1405 — does the prediction interval hold beyond the six primary seeds?
- Sup. C: Analysis Provenance — evidentiary role and multiplicity handling of each analysis. (`sec:provenance`)
- Sup. D: Canonical Checkpoint Registry — what was trained, and which families enter which claim? (`tab:checkpoint_registry`)
- Sup. E: Donor-Pool Substitution Decomposition — how much of the seen-pool advantage is pool-origin vs. similarity? (`sec:appendix_decomposition`, `tab:path_sensitivity`)
- Sup. F: Config-Faithful and Perturbation Details — protocol-mismatch check, weight-noise sweep, epoch-checkpoint decoupling search. (`sec:appendix_config_faithful`, `sec:appendix_perturbation`, `tab:perturbation`)
- Sup. G: Cluster-Aware Bootstrap — does the gap survive resampling by signer, broadcast-show, and date? (`sec:appendix_cluster_bootstrap`)
- Sup. H: Supplementary Oracle Diagnostics — is the evaluator-aware advantage robust to reference-overlap exclusion? (`sec:appendix_oracle`)
- Sup. I: Template-Density, Holdout Boundary, Donor-Exclusion — does the reversal persist on a clean template-holdout split and under donor-text exclusion? (`sec:appendix_template_boundary`)
- Sup. J: Transcript-Slot Diagnostic Details — do slot-level scores reproduce the BLEU reversal? (`sec:appendix_slot`)
- Sup. K: Competence Extrapolation and Per-Item Gap Decomposition — could competence in the unobserved interval explain the reversal, and which item features drive the gaps? (`sec:appendix_extrapolation`)
- Sup. L: CSL-Daily Boundary Control — does the readout signature generalize beyond PHX-public? (`sec:csl-daily`)
- Sup. M: Leakage Sanity Checks — can the train-pool readout be explained by implementation shortcuts? (`sec:appendix_leakage`)
- Sup. N: Pre-Exclusion Materialization Sensitivity — how sensitive are the headline numbers to the donor-exclusion rule? (`sec:appendix_materialization`)
- Sup. O: BLEURT-Substitute Validation — does a learned metric corroborate the BLEU finding under reference replacement? (`sec:appendix_bleurt`)
- Sup. P: Main-Text Tables (SI Table S22–S31) — decoded-metric sensitivity, floor effect, cross-metric gaps, leaderboard, distillation ladder, checkpoint registry, protocol summary, rescue sweep, exposure estimands, reference sensitivity. (`sec:appendix_moved_tables`)
- Sup. Q: Additional Protocol Details — tie-breaking, scorer equivalence, metric implementations, audit-design, reference-frame, and donor-origin figures (SI Fig. S4–S6). (`sec:appendix_moved_protocol`)

Full checkpoint registry in SI Sup. D.

References

Alkain, B., A. Núñez-Marcos, C. Escolano, L. Docío-Fernández, O. Perez-de Viñaspre, and G. Labaka. 2026. Critical analysis of datasets for sign language translation. *Frontiers in Artificial Intelligence* 9: 1743223. https://doi.org/10.3389/frai.2026.1743223 .

Artiaga, K., S. Kamila, H. Afli, C. Lynch, and M. Hasanuzzaman 2025. Rethinking sign language translation: The impact of signer dependence on model evaluation. In

Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China, pp. 18379–18391. Association for Computational Linguistics.

Camgoz, N.C., S. Hadfield, O. Koller, H. Ney, and R. Bowden 2018. Neural sign language translation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7784–7793.

Carlini, N., F. Tramer, E. Wallace, M. Jagielski, A. Herbert-Voss, K. Lee, A. Roberts, T. Brown, D. Song, U. Erlingsson, A. Oprea, and C. Raffel 2021. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pp. 2633–2650.

Cory, O., M. Ivashechkin, K. Sahin, O. Ranum, J. Low, E. Fish, A. Pelykh, et al. 2025. BackTranslation2.0: A linguistically motivated metric to assess sign language production. *arXiv preprint arXiv:2606.28673*. arXiv:2606.28673 .

Czehmann, V., S. Yazdani, Y. Hamidullah, F. Nunnari, and E. Avramidis 2025. "A Sacred Bird Called the Phoenix": Auditing the most-used parallel corpus for German sign language recognition and translation. In *Proceedings of the LREC2026 12th Workshop on the Representation and Processing of Sign Languages: Language in Motion*, Palma, Mallorca, Spain, pp. 80–92. ELRA Language Resources Association. ISBN 978-2-493814-82-1. Back-translated test set and annotations: https://github.com/DFKI-SignLanguage/sacre-bird-phoenix (CC BY-NC-SA 4.0).

He, T., K. Rahul, D. Hupkes, and R. Cotterell 2024. On the blind spots of model-based evaluation metrics for text generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*.

Hong, R. and Košecká, Jana. 2025. Conditional collapse in sign language production: A diagnostic and a scaling argument. *arXiv preprint arXiv:2606.01643*. arXiv:2606.01643 [cs.CV].

Imai, S. 2025. SiLVERScore: Why aren't semantically-aware embeddings used for sign language generation evaluation? In *Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2025)*, pp. 519–528.

Jiang, Z. et al. 2025. Meaningful pose-based sign language evaluation. In *Proceedings of the 10th Workshop on Machine Translation (WMT)*.

Kandpal, N., E. Wallace, and C. Raffel 2022. Looking inside my memory: Quantifying memorization in language models. In *Findings of EMNLP*.

Khandelwal, U., A. Fan, D. Jurafsky, L. Zettlemoyer, and M. Lewis 2020. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations (ICLR)*.

Kim, J.H., M. Huerta-Enochian, C. Ko, and D.H. Lee 2024, May. SignBLEU: Automatic evaluation of multi-channel sign language translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Torino, Italia, pp. 14796–14811. ELRA and ICCL.

Post, M. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pp. 186–191.

Saunders, B., N.C. Camgoz, and R. Bowden 2020. Progressive transformers for end-to-end sign language production. In *European Conference on Computer Vision*.

Saunders, B., N.C. Camgöz, and R. Bowden. 2021. Continuous 3D multi-channel sign language production via progressive transformers and mixture density networks. *International Journal of Computer Vision* 129: 2113–2135. https://doi.org/10.1007/s11263-021-01457-9 .

Tasyurek, S.M., T. Kiziltepe, and H.Y. Keles. 2025. Disentangle and regularize: Sign language production with articulator-based disentanglement and channel-aware regularization. *arXiv preprint arXiv:2504.06610*. arXiv:2504.06610 .

Walsh, H., E. Fish, O. Mercanoglu Sincan, M.I. Lakhal, R. Bowden, N. Fox, B. Woll, K. Wu, Z. Li, W. Zhao, H. Wang, W. Zhou, H. Li, S. Tang, J. He, X. Wang, R. Zhang, Y. Wang, L. Cheng, M. Tasyurek, T. Kiziltepe, and H.Y. Keles 2025. SLRTP2025 sign language production challenge: Methodology, results, and future work. In *Proceedings of the Third Sign Language Recognition, Translation and Production Workshop at CVPR 2025*. arXiv:2508.06951.

Walsh, H., A. Ravanshad, M. Rahmani, and R. Bowden 2024. A data-driven representation for sign language production. In *British Machine Vision Conference*.

Walsh, H., B. Saunders, and R. Bowden 2024. Sign stitching: A novel approach to sign language production. In *British Machine Vision Conference*.

Yazdani, M. et al. 2025. A critical study of automatic evaluation in sign language translation. In *Proceedings of LREC-COLING 2026*.

Zhang, T., V. Kishore, F. Wu, K.Q. Weinberger, and Y. Artzi 2020. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR)*.

Zuo, R., R.A. Potamias, E. Ververas, J. Deng, and S. Zafeiriou 2025. Signs as tokens: A retrieval-enhanced multilingual sign language generator. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.